

Our Project, but with Functors: (possible) Answers

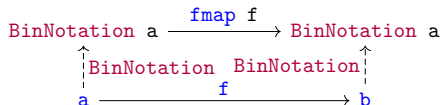
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December 1, 2025

- Natural numbers on binary notations:
 - functors to model them
 - unary operations

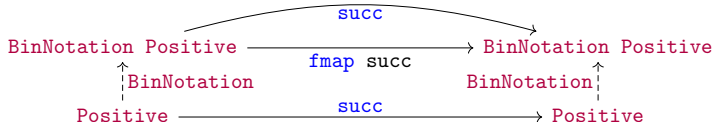
Binary built on Positive by means of a functor

Define



and implement it in `Numbers.Binary.Functor.Structure.hs`

Take the instance



implement it in `Numbers.Binary.Functor.OperationsUnary.hs`

Exercises

Alternatives to the representation by BN

- Focus on `Numbers.Positive.Exercises.PList.hs`

Extend it to represent natural numbers in binary, looking at

`Numbers.Binary.Functor.Exercises.BNList`

Solutions:

- `Numbers.Positive.Exercises.PList_SOLUTION` (restricted)
- `Numbers.Binary.Functor.Exercises.BNList_SOLUTION` (restricted)

- Focus on `Numbers.Positive.Exercises.PEither.hs`

Extend it to represent natural numbers in binary, looking at

`Numbers.Binary.Functor.Exercises.BNEither.hs`

Solutions:

- `Numbers.Positive.Exercises.PEither_SOLUTION` (restricted)
- `Numbers.Binary.Functor.Exercises.BNEither_SOLUTION` (restricted)



Milewski, B. and I. Tabachnik (2019). *Category Theory for Programmers*.

<https://bartoszmilewski.com/2014/10/28/category-theory-for-programmers-the-preface/>. Lightning Source UK. ISBN: 9780464243878. URL:

<https://books.google.it/books?id=hA55yQEACAAJ>.